

Exact minimum measurement dependence for violating the CHSH inequality

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We compute the exact minimum measurement dependence that a local hidden-variable model needs in order to reproduce the maximal quantum violation of the CHSH inequality, measured as the worst-case total-variation distance $F = \max_{ab} \text{TV}(p(\cdot | a, b) \| \rho)$ between the settings-conditioned hidden-variable distribution and a fixed baseline ρ . Every local model — on any hidden-variable space, with any source distribution and deterministic or stochastic responses — obeys $S \leq 2 + 8F$, so reaching the Tsirelson value $2\sqrt{2}$ costs at least $F^* = (\sqrt{2} - 1)/4 \approx 0.104$. This floor is attained by an explicit four-state model, verified as an exact identity in $\mathbb{Q}(\sqrt{2})$, and adding hidden states beyond four never lowers the cost: for N -state uniform-source models $F^*(N) = (\sqrt{2} - 1)/\min(N, 4)$. The same minimum is achievable without observable signalling for every $N \geq 3$, with explicit no-signalling witnesses, while no two-state model can violate Bell under no-signalling at any fine-tuning level. We further give the exact piecewise-linear curves $S_{\max}(N, F)$ for $N \leq 14$ by exhaustive exact arithmetic, the cost $\lceil N/4 \rceil / N$ of reaching the algebraic maximum $S = 4$ with or without no-signalling (where the unique no-signalling box is the PR box), the one-sided value $(\sqrt{2} - 1)/2$, and conversion bounds to Hall’s measure and to mutual information. The qualitative lesson that modest measurement dependence suffices is known; the contribution is an exact, certified, model-independent price in one natural metric, together with the observation that it can be paid without leaving a trace in the observable marginals.

I. INTRODUCTION

Bell’s theorem [1], in the Clauser–Horne–Shimony–Holt (CHSH) form [2], is the cleanest statement of quantum nonlocality: every local hidden-variable model satisfies $|S| \leq 2$, while quantum theory reaches the Tsirelson bound $S = 2\sqrt{2}$ [3] and the algebraic maximum is 4. The theorem rests on two assumptions — locality (factorisability) and statistical independence of the settings from the hidden variable (“free choice”). Relaxing either one evades the theorem. The first assumption itself decomposes into parameter independence and outcome independence [4, 5], and local hidden-variable models are equivalently characterised by the existence of a joint probability for all four outcomes [6]. This paper quantifies the second escape route: *how much* may a local model’s hidden variable depend on the measurement settings, and what is the minimum amount required to reproduce a given CHSH value?

The question converts the qualitative objection “superdeterministic models are conspiratorial” into a quantitative benchmark. If the required fine-tuning is small, the loophole is cheap and Bell’s theorem says little about the real world; if it is non-trivial, every local explanation must bite hard. Earlier work established that only a modest relaxation suffices. Brans showed already in 1988 that Bell’s theorem does not eliminate fully causal — i.e. measurement-dependent — hidden-variable models [7]; the resulting measurement-dependence (“superdeterminism”) programme was developed by the later works below and reappraised recently in Ref. 8. Hall showed that a local deterministic model of singlet-state correlations can be built by giving up a fraction of measurement independence, with a cost of $M_{\text{singlet}} = 2(\sqrt{2} - 1)/3 \approx 0.276$ in his pairwise L^1 metric M [9]. Barrett and Gisin showed that all projective singlet correlations can be reproduced

with Bob’s settings fully free and at most one bit of mutual information between Alice’s setting and the hidden variable, and related such models to communication and detection-loophole models [10]; the consequences for Bell-based randomness expansion were analysed in Ref. 11. Pütz *et al.* characterised the measurement-dependent local correlations as a polytope and showed that, in their parametrisation by bounds on the conditional setting probabilities, an arbitrarily small amount of residual measurement independence suffices to manifest quantum nonlocality for suitable correlations [12, 13]. Friedman *et al.* derived tight relaxed CHSH inequalities for arbitrary per-observer measurement dependence [14]; Hall and Branciard quantified the cost in mutual information bits under different causal structures [15]. More recent work has developed “exact minimum measurement dependence” programs in Hall-type metrics, including trade-offs with hidden information [16] and, most recently, exact closed forms for CHSH and GHZ–Mermin inequalities [17, 18]; a total-variation-based bound for drifting preparation ensembles has also appeared [19]. Bacciagaluppi, Hermens and Leegwater argued that some measurement-independence violations amount to “signalling in principle” and that Bell’s theorem holds if no-signalling is imposed [20].

We work in a complementary metric. For a model with hidden variable λ and conditional sources $p(\lambda | a, b)$, we define the fine-tuning cost as the worst-case total-variation distance of any conditional source from a fixed baseline source ρ ,

$$F = \max_{ab} \text{TV}(p(\cdot | a, b) \| \rho),$$

$$\text{TV}(p \| \rho) = \frac{1}{2} \sum_{\lambda} |p(\lambda | a, b) - \rho(\lambda)|, \quad (1)$$

$F = 0$ is exactly Bell’s statistical-independence assumption; F measures, per settings block, how far the source

must be bent toward the settings. The metric is deliberately not information-theoretic: it is a direct measure of the deviation of the joint distribution $p(\lambda, a, b)$ from the product form, in the same space where Bell’s assumptions live.

Our main results are:

1. **A model-independent bound.** For every local CHSH model — on any hidden-variable space, with any source distribution and deterministic or stochastic responses — $S \leq 2 + 8F$. The bound is tight on the interval $[0, 1/N]$ for every $N \geq 4$ (Sec. IV A).
2. **Exact curves.** For N -state models with a uniform source, $S_{\max}(N, F)$ is an exact piecewise-linear function of F , given in closed form for $N = 2, \dots, 8$ by exhaustive exact arithmetic; the cap $F(S=4; N) = \lceil N/4 \rceil / N$ holds for all N (Secs. IV B and IV D).
3. **The minimum fine-tuning for the Tsirelson value.** For uniform-source models, $F^*(N) := \min\{F : S_{\max}(N, F) \geq 2\sqrt{2}\} = (\sqrt{2} - 1) / \min(N, 4)$: 0.20711 at $N = 2$, 0.13807 at $N = 3$, and $(\sqrt{2} - 1)/4 \approx 0.103553$ for all $N \geq 4$. The cost *saturates* at $N = 4$: more hidden states do not make the loophole cheaper; and the floor $F^* \geq (\sqrt{2} - 1)/4$ is model-independent, holding for every local model (Secs. IV A and IV C).
4. **No-signalling is silent.** Imposing full no-signalling on the observed marginals does not change the minimum for any $N \geq 3$: $F_{\text{NS}}^*(N) = F^*(N)$, with explicit algebraic witnesses at every N ; the same silence holds at the algebraic maximum, $F_{\text{NS}}^*(S=4; N) = \lceil N/4 \rceil / N$, the signalling-allowed cap, since the unique NS box with $S = 4$ is the PR box (Sec. V D); and at $N = 2$, no fine-tuning level permits a Bell violation under no-signalling (Sec. V).
5. **An explicit witness.** A four-state model attaining $S = 2\sqrt{2}$ at exactly $F^* = (\sqrt{2} - 1)/4$, verified as an exact identity in $\mathbb{Q}(\sqrt{2})$ (Sec. VI).

We are careful about the strength of these claims. This is not a refutation of Bell’s theorem, and we do not claim it as a breakthrough: the qualitative statement that “only a little fine-tuning suffices” is already in the literature [9, 14, 15]. What is new — no prior use was found in arXiv-indexed literature through 2026-09-02, a completed sweep we describe in Sec. VII C — is (i) the exact closed form $F^* = (\sqrt{2} - 1) / \min(N, 4)$ in the per-setting TV metric, with the $N = 4$ saturation, the exact curves and cap, and the model-independent bound $S \leq 2 + 8F$; (ii) the no-signalling sub-case pinned to the same closed form for all $N \geq 3$, with explicit witnesses, plus the sharp two-state impossibility theorem and the algebraic-maximum cap $F_{\text{NS}}^*(S=4; N) = \lceil N/4 \rceil / N$ (the

PR box is the unique NS box with $S = 4$) — a precise quantitative answer, in the observable-marginal sense, to the “signalling in principle” question of Ref. 20; (iii) the explicit finite witness at exactly F^* ; and (iv) the one-sided causal-structure value $F_{\text{os}}^* = (\sqrt{2} - 1)/2$, flat in N , and the explicit metric-conversion bounds of Sec. VII E.

II. MODEL AND DEFINITIONS

A. The finite model

Fix the CHSH scenario: settings $a, b \in \{0, 1\}$, outcomes $x, y \in \{\pm 1\}$, correlators $E[ab] = \langle xy \rangle_{a,b}$, and

$$S = E[00] + E[01] + E[10] - E[11] = \sum_{ab} \sigma_{ab} E[ab], \quad (2)$$

with $\sigma_{ab} = +1$ for $ab = 00, 01, 10$ and $\sigma_{11} = -1$, the ab being ordered as 00, 01, 10, 11. A *finite local model* with N hidden states is specified by:

- a uniform source $\rho(\lambda) = 1/N$ on $\lambda = 1, \dots, N$;
- for each joint setting (a, b) , a *conditional source* $p[ab, \lambda] \geq 0$ with row sum $\sum_{\lambda} p[ab, \lambda] = 1$. The settings may bias which source states occur — this is the measurement dependence;
- deterministic responses $u[a, \lambda], v[b, \lambda] \in \{\pm 1\}$ for Alice and Bob.

The correlators are $E[ab] = \sum_{\lambda} p[ab, \lambda] u[a, \lambda] v[b, \lambda]$, and the fine-tuning cost is Eq. (1). We write

$$S_{\max}(N, F) = \sup S \quad (3)$$

over all such models with fine-tuning at most F .

We also study the *no-signalling (NS) sub-case*: the observed marginals must not depend on the distant setting, i.e. for every a , $\sum_{\lambda} p[ab, \lambda] u[a, \lambda]$ is independent of b , and symmetrically for Bob. This asks whether the fine-tuning must leak into observable signalling, or whether a local model can hide it.

B. Product patterns and the parity lemma

For each λ define the *product pattern* $q_{\lambda} = (q_{00}, q_{01}, q_{10}, q_{11})$ with $q_{ab}(\lambda) = u[a, \lambda]v[b, \lambda]$. The product telescopes,

$$q_{00} q_{01} q_{10} q_{11} = (u[0, \lambda]u[1, \lambda])(v[0, \lambda]v[1, \lambda]) = +1, \quad (4)$$

so only the 8 *even-parity* patterns occur. Each has CHSH value $C(q) := q_{00} + q_{01} + q_{10} - q_{11} \in \{\pm 2\}$: four patterns have $C = +2$ — the *miss-type* patterns, each of which differs from σ at exactly one coordinate —

$$\begin{aligned} M_A &= (-1, +1, +1, -1), & M_B &= (+1, -1, +1, -1), \\ M_C &= (+1, +1, -1, -1), & M_D &= (+1, +1, +1, +1). \end{aligned} \quad (5)$$

and the other four are their negatives, with $C = -2$. The sign vector $\sigma = (+, +, +, -)$ itself has product -1 , so *no even-parity pattern equals σ* : every state misses at least one of the four conditions $q_{ab} = \sigma_{ab}$ that would give $S = 4$. We call this the *parity lemma*; it is the key combinatorial fact behind the cap theorem (Sec. IV B).

III. EXACT REDUCTION TO A FINITE MAXIMUM

Two structural facts make $S_{\max}(N, F)$ exactly computable, and the exact statements below rest on them rather than on any numerical solver.

A. The per-row TV optimum

Lemma 1 (per-row TV optimum). *Let $f \in \{\pm 1\}^N$ have k entries equal to $+1$ and m entries equal to -1 ($k + m = N$), set $\mu = (k - m)/N$, and let $\mathcal{D}_F = \{p \in \Delta_N : \text{TV}(p||\rho) \leq F\}$ with ρ uniform. Then*

$$\begin{aligned} \max_{p \in \mathcal{D}_F} p \cdot f &= \mu + 2 \min(F, m/N), \\ \min_{p \in \mathcal{D}_F} p \cdot f &= \mu - 2 \min(F, k/N), \end{aligned} \quad (6)$$

with both sides equal to μ in the degenerate cases $k = 0$ or $m = 0$.

Proof. Write $d = p - \rho$ (zero-sum). Then $p \cdot f - \mu = d \cdot f$, and two upper bounds apply. (*TV bound*) Since $|f_\lambda| = 1$, $|d \cdot f| \leq \sum_\lambda |d_\lambda| = 2 \text{TV}(p||\rho) \leq 2F$. (*Mass bound*) Let $M^- = \sum_{f=-1} d_\lambda$; since $p \geq 0$, each of the m terms is $\geq -1/N$, so $M^- \geq -m/N$. Using $\sum_\lambda d_\lambda = 0$, $d \cdot f = (\sum_{f=+1} d_\lambda) - M^- = -2M^- \leq 2m/N$; symmetrically $d \cdot f \geq -2k/N$. Combining gives the upper bound. Achievability is bang-bang: put all of $+\delta$ on one $+1$ entry ($p = 1/N + \delta$ there, legal since $\delta \leq m/N \leq (N-1)/N$) and spread $-\delta$ over $\lceil \delta N \rceil$ entries of the -1 set (each can absorb at most $1/N$, and $\lceil \delta N \rceil \leq m$ for $\delta \leq m/N$). Then $d \cdot f = 2\delta$ exactly with $\text{TV}(p||\rho) = \delta$. The minimum follows by moving mass from the $+1$ set to the -1 set, where the available mass is k/N . $\square$

Lemma 1 says: a row can only be bent toward f at rate 2 per unit of TV, and it saturates once F reaches the fraction m/N of states that already point the “wrong” way. It is the only non-combinatorial ingredient in this paper; everything else is finite combinatorics on top of it.

B. The exact finite-maximum formula

Theorem 1 (exact finite maximum). *For all $N \geq 2$ and all $F \in [0, 1]$,*

$$S_{\max}(N, F) = \max_Q \left[S_0(Q) + \sum_{ab} g_{ab}(F) \right], \quad (7)$$

where Q ranges over multisets of N even-parity patterns (from the eight of Sec. II), $S_0(Q) = \frac{1}{N} \sum_{\lambda \in Q} C(q_\lambda)$, $n_{ab}(Q) = \#\{\lambda \in Q : q_{ab}(\lambda) \neq \sigma_{ab}\}$, and

$$g_{ab}(F) = \begin{cases} 2 \min(F, n_{ab}/N), & 0 < n_{ab} < N, \\ 0, & n_{ab} \in \{0, N\}. \end{cases} \quad (8)$$

There is no σ_{ab} sign in front of the min-terms: each row ab is optimised in its own σ_{ab} -direction (toward $E[ab] = \sigma_{ab}$), so every contribution to S is positive.

Proof sketch. Four steps. **(i) Vertices.** For fixed p , S is multi-affine in the response variables $u[a, \lambda], v[b, \lambda]$ (linear in each, since only products enter); its maximum over $\{\pm 1\}^{4N}$ equals its maximum over $[-1, 1]^{4N}$ and is attained at a vertex. Swapping the two maxima — legal because the p -feasible set (product of per-row TV balls with the simplex) is independent of (u, v) , compact, and S is continuous in both factors, so the maximum over the product space is attained and the two orders agree — gives $S_{\max}(N, F) = \max_{(u,v) \in \{\pm 1\}^{4N}} \text{LP}(u, v)$, where LP maximises S over p with row sums 1 and per-row TV $\leq F$.

(ii) Gauge reduction. S depends on (u, v) only through the products $q_{ab}(\lambda)$: $E[ab] = \sum_\lambda p[ab, \lambda] q_{ab}(\lambda)$. The per-state simultaneous sign flip $(u[\cdot, \lambda], v[\cdot, \lambda]) \mapsto (-u[\cdot, \lambda], -v[\cdot, \lambda])$ leaves all q 's fixed, and any two vertices with the same pattern assignment have identical LP values. The maximum over 2^{4N} vertices reduces to a maximum over pattern assignments $Q = (q_1, \dots, q_N)$, each even-parity.

(iii) Realisability. Every even-parity multiset is realisable by some vertex: given q_λ with product $+1$, set $u[0, \lambda] = +1$, $v[0, \lambda] = q_{00}$, $v[1, \lambda] = q_{01}$, $u[1, \lambda] = q_{00}q_{10}$; then $u[1, \lambda]v[0, \lambda] = q_{10}$ and $u[1, \lambda]v[1, \lambda] = q_{00}q_{01}q_{10} = q_{11}$ by even parity.

(iv) Row separability. For fixed Q , the LP objective $\sum_{ab} \sigma_{ab} \sum_\lambda p[ab, \lambda] q_{ab}(\lambda)$ and all constraints are separable in ab : each block $\{p[ab, \cdot]\}_\lambda$ is an independent probability vector subject to its own TV ball. Apply Lemma 1 to row ab with target direction $f_\lambda = \sigma_{ab} q_{ab}(\lambda)$; the “wrong-sign” count of Lemma 1 is then exactly $n_{ab}(Q)$ and $\mu_{ab} = \frac{1}{N} \sum_\lambda \sigma_{ab} q_{ab}(\lambda)$, giving $\max_{\text{row } ab} \sigma_{ab} \sum_\lambda p[ab, \lambda] q_{ab}(\lambda) = \mu_{ab} + g_{ab}(F)$. Summing over ab and using $\sum_{ab} \mu_{ab} = S_0(Q)$ gives Eq. (7). $\square$

Consequences. For each N , $S_{\max}(N, F)$ is an *exact* piecewise-linear function of F : on each cell $F \in [k/N, (k+1)/N]$ every term $2 \min(F, n/N)$ is affine, so $S_{\max}$ is the upper envelope of finitely many exact affine functions, computable in rational arithmetic over the $\binom{8+N-1}{N}$ multisets (36/120/330/792/1716/3432/6435 for $N = 2, \dots, 8$). The slopes can only be 0, 2, 4, 6, 8.

C. The round-robin construction

Let Q_N be the multiset formed by the first N entries of the periodic sequence $M_A, M_B, M_C, M_D, M_A, M_B, \dots$, i.e. state i takes miss-type $M_{i \bmod 4}$. Then $S_0(Q_N) = 2$ (all patterns have $C = +2$) and each count $n_{ab}(Q_N)$ is $\lfloor N/4 \rfloor$ or $\lceil N/4 \rceil$ (the four counts differ by at most one). The construction attains

$$S = 2 + 8F \quad \text{on } [0, 1/N] \text{ for every } N \geq 4, \quad (9)$$

since on that interval $F \leq 1/N \leq \lfloor N/4 \rfloor / N$ and all four rows are active. It is the single explicit family of response tables that gives the entire lower-bound curve for every $N = 2, \dots, 8$ (Sec. IV D).

IV. RESULTS WITH SIGNALLING ALLOWED

A. The global bound and its generality

Theorem 2. *For every $N \geq 2$ and every model,*

$$S \leq 2 + 8F, \quad (10)$$

independently of N . The bound is tight on $[0, 1/N]$ for every $N \geq 4$.

Proof. For any model the identity $S = S_0 + \sum_{ab} (\sigma_{ab} E[ab] - \mu_{ab})$ holds, with μ_{ab} as in step (iv) of Theorem 1: summing its definition over ab gives $\sum_{ab} \mu_{ab} = \frac{1}{N} \sum_{\lambda} C(q_{\lambda}) = S_0$. Lemma 1 gives $\sigma_{ab} E[ab] - \mu_{ab} \leq 2 \min(F, n_{ab}/N) \leq 2F$ per row, so $S \leq S_0 + 8F$. Finally $S_0 = \frac{1}{N} \sum_{\lambda} C(q_{\lambda}) \leq 2$, since every even-parity pattern has $C(q) \in \{\pm 2\}$. Tightness on $[0, 1/N]$ for $N \geq 4$ is Eq. (9). $\square$

The bound holds far beyond the finite uniform-source setting:

Proposition 1 (general-measure extension). *Inequality (10) holds for any local CHSH model on an arbitrary probability space (Λ, ρ) , with any source distribution ρ (uniformity is not used) and with deterministic or stochastic responses, where $F = \max_{ab} \text{TV}(p(\cdot | a, b) || \rho)$.*

Proof sketch. (i) *Deterministic responses.* Pointwise, q_{λ} has even parity, so $C(q_{\lambda}) \in \{\pm 2\}$ almost everywhere and $S_0 := \sum_{ab} \mu_{ab} = \int C(q) d\rho \leq 2$, where $\mu_{ab} := \int \sigma_{ab} q_{ab} d\rho$ is the measure-theoretic analogue of the finite-model definition. With ν_{ab} denoting the conditional source measure, $\nu_{ab}(\{\lambda\}) = p(\lambda | ab)$, the identity $\sigma_{ab} E[ab] - \mu_{ab} = \int \sigma_{ab} q_{ab} d(\nu_{ab} - \rho)$ holds, and for any f with $\|f\|_{\infty} \leq 1$, $|\int f d(\nu - \rho)| \leq \int |f| d|\nu - \rho| \leq 2 \text{TV}(\nu || \rho)$; hence $|\sigma_{ab} E[ab] - \mu_{ab}| \leq 2F$ per row, and since $S = S_0 + \sum_{ab} (\sigma_{ab} E[ab] - \mu_{ab})$, we get $S \leq S_0 + 8F \leq 2 + 8F$.

(ii) *Stochastic responses.* Given λ , the four response distributions define a joint measure μ_{λ} on $\{\pm 1\}^4$ (local response independence given λ). The model equals

the *deterministic* model on $\Lambda \times \{\pm 1\}^4$ with source $\rho'(\lambda, s) = \rho(\lambda) \mu_{\lambda}(s)$, conditional sources $p'_{ab}(\lambda, s) = p_{ab}(\lambda) \mu_{\lambda}(s)$, and responses s : the correlators match because $\sum_s \mu_{\lambda}(s) s_a t_b = \alpha_a(\lambda) \beta_b(\lambda)$ with $\alpha_a = E[x | a, \lambda]$, $\beta_b = E[y | b, \lambda]$. The TV cost is preserved *exactly*, since μ_{λ} factors out of the TV sum: $\frac{1}{2} \sum_{\lambda, s} |p'_{ab} - \rho'| = \frac{1}{2} \sum_{\lambda} |p_{ab}(\lambda) - \rho(\lambda)| \sum_s \mu_{\lambda}(s) = \text{TV}(p_{ab} || \rho)$. Case (i) applies to the enlarged model. $\square$

Consequence. The lower bound $F^* \geq (\sqrt{2} - 1)/4$ derived below is a statement about *all* local models — every source distribution, deterministic or stochastic — not just finite uniform-source ones. No extension of the model class can lower the fine-tuning floor.

B. The cap: reaching $S = 4$

Theorem 3. *For all $N \geq 2$, the smallest fine-tuning that lets a local deterministic model reach $S = 4$ is*

$$F(S=4; N) = \frac{\lceil N/4 \rceil}{N}. \quad (11)$$

For $N = 2, \dots, 8$ the values are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{2}{5}, \frac{1}{3}, \frac{2}{7}, \frac{1}{4}$.

Proof. Since $|E[ab]| \leq 1$, $S = \sum_{ab} \sigma_{ab} E[ab] \leq \sum_{ab} |E[ab]| \leq 4$, with equality only if $E[ab] = \sigma_{ab}$ for all four ab . By Lemma 1, row ab can reach σ_{ab} only if $F \geq n_{ab}/N$ (for $0 < n_{ab} < N$; $n_{ab} = 0$ needs no budget), so $S = 4$ is achievable at budget F iff some Q has $n_{ab} \leq FN$ for all ab , i.e. $F(S=4; N) = \min_Q \max_{ab} n_{ab}/N$. By the parity lemma each λ misses at least one class, so $\sum_{ab} n_{ab} \geq N$ and hence $\max_{ab} n_{ab} \geq \lceil N/4 \rceil$ (necessity). The round-robin construction Q_N has $\max_{ab} n_{ab} = \lceil N/4 \rceil$, and at $F = \lceil N/4 \rceil / N$ every row can be supported entirely on its correct class ($\text{TV} = n_{ab}/N \leq F$), giving $S = 4$ (sufficiency). $\square$

Note that the cap is *not* “ $1/4$ for all $N \geq 4$ ”; e.g. $S_{\max}(5, \frac{1}{4}) = \frac{16}{5} + \frac{1}{2} = 3.7 < 4$. The frequently quoted statement “ $S = 4$ requires $F \geq \frac{1}{4}$ ” remains true ($\lceil N/4 \rceil / N \geq \frac{1}{4}$ always) but is tight only when $4 | N$. The non-monotonicity in N is a property of the *uniform-baseline* class, in which every one of the N states carries baseline mass $1/N$ and none can be left unused; with an arbitrary baseline ρ the $N = 4$ model embeds into any $N \geq 4$, and the cap is $1/4$ for all $N \geq 4$, in agreement with Theorem 2.

C. The minimum fine-tuning for the Tsirelson value

Theorem 4 (headline). *For all $N \geq 2$,*

$$F^*(N) := \min\{F : S_{\max}(N, F) \geq 2\sqrt{2}\} = \frac{\sqrt{2} - 1}{\min(N, 4)}. \quad (12)$$

is non-constant (e.g. $u[0, \cdot] \neq \text{constant} \Rightarrow t_{00} = t_{01}$). The four NS equations therefore collapse the four free variables to a small number of independent groups, and S is an affine function of those groups over a box of half-width F . With only one side constrained, half these pinning relations are absent and a slope-4 direction survives (the side finding below). Enumerating all $2^8 = 256$ response vertices and closing under the four NS equations by union-find reduces the free variables to groups; maximising the resulting affine S over the box gives, for each vertex, an explicit exact affine function $A + BF$ (the sign of B fixes which endpoint is maximal). The enumeration produces 1312 candidate affine functions $A + BF$ in total; every one satisfies $A + BF \leq 2$ for all $F \in [0, \frac{1}{2}]$ (checked at both endpoints), and $S = 2$ is attained for every $F \in [0, \frac{1}{2}]$ (e.g. the constant responses $u \equiv v \equiv +1$, for which $E[ab] = 1$ and hence $S = 2$ at every feasible row). Hence $S_{\text{NS}}(2, F) \equiv 2$. The certificate is exact (rational arithmetic) and re-runnable; see Appendix C. $\square$

Side finding. Under *one-sided* NS only (Alice's marginals independent of Bob's setting), the bound is not flat — at $N = 2$, $S = 2 + 4F$ is reachable; the flatness genuinely uses both sides. (This is a constraint on the observed marginals; the different constraint that the hidden source itself be independent of Bob's setting — one-sided *measurement dependence* — is treated separately in Sec. VII B.)

C. All $N \geq 3$: no-signalling is silent

Theorem 7 (NS headline). *Under full no-signalling, for every $N \geq 3$,*

$$\begin{aligned} F_{\text{NS}}^*(N) &:= \min\{F : S_{\text{NS}}(N, F) \geq 2\sqrt{2}\} \\ &= \frac{\sqrt{2} - 1}{\min(N, 4)} = F^*(N). \end{aligned} \quad (14)$$

Fine-tuning to the CHSH Tsirelson value is completely NS-silent: the minimum is identical with and without signalling, and it is attained by an explicit no-signalling model at every $N \geq 3$.

Proof. Lower bound. NS restricts the feasible set, so $S_{\text{NS}}(N, F) \leq S_{\text{max}}(N, F)$. For $N \geq 4$, Theorem 2 gives $S_{\text{max}} \leq 2 + 8F < 2\sqrt{2}$ whenever $F < (\sqrt{2} - 1)/4$; for $N = 3$, the exact curve of Theorem 5, $S_{\text{max}}(3, F) = 2 + 6 \min(F, \frac{1}{3})$, gives $S_{\text{max}} < 2\sqrt{2}$ whenever $F < (\sqrt{2} - 1)/3$. Hence $S_{\text{NS}}(N, F) < 2\sqrt{2}$ for all $N \geq 3$ with $F < (\sqrt{2} - 1)/\min(N, 4)$.

Upper bound — explicit algebraic witnesses.

$N \geq 4$. Let $d = F^* = (\sqrt{2} - 1)/4$. State $i = 0, \dots, N - 1$ lies in class $j = i \bmod 4$, with class counts $n_j \in \{\lfloor N/4 \rfloor, \lceil N/4 \rceil\}$. The construction is:

- *Patterns:* state i has miss-type M_j (all four $C = +2$ patterns; Sec. II).

- *Response gauge* (per class j):

$$\begin{aligned} u[0, \cdot] &= (+, +, -, -), & u[1, \cdot] &= (-, +, +, -), \\ v[0, \cdot] &= (-, +, -, -), & v[1, \cdot] &= (+, -, -, -), \end{aligned} \quad (15)$$

i.e. the j -th entry of each tuple gives the response of states in class j . One checks that the resulting per-state products are exactly M_A, M_B, M_C, M_D for classes 0, 1, 2, 3.

- *Rows:* $p[ab, i] = \frac{1}{N} + x[ab, i \bmod 4]$ with (all other $x = 0$)

$$\begin{aligned} x[00, 0] &= -d/n_0, & x[00, 1] &= +d/n_1, \\ x[01, 0] &= +d/n_0, & x[01, 1] &= -d/n_1, \\ x[10, 1] &= +d/n_1, & x[10, 2] &= -d/n_2, \\ x[11, 0] &= +d/n_0, & x[11, 3] &= -d/n_3. \end{aligned} \quad (16)$$

Verification is exact (rational and $\mathbb{Q}(\sqrt{2})$ arithmetic; certificate in Appendix C):

- *Rows.* $n_r x[ab, r] = -d$ and $\sum_{j \neq r} n_j x[ab, j] = +d$ hold identically for each row, so every row has TV exactly d and is bang-bang in its σ_{ab} -direction. Feasibility $p \geq 0$ requires $d/n_r \leq 1/N \iff n_r \geq dN$ for all r — precisely the case-checked inequality $\lfloor N/4 \rfloor \geq (\sqrt{2} - 1)N/4$ of Sec. IV C.
- *NS.* Each NS equation — Alice's equating rows $a0$ and $a1$ (weighted by u_a), Bob's equating rows $0b$ and $1b$ (weighted by v_b) — reduces to $\sum_j n_j x[ab_L, j] w_j = \sum_j n_j x[ab_R, j] w_j$ for the response values w_j (the uniform parts cancel). Substituting Eqs. (15) and (16), each side is a sum of at most two terms; all four close explicitly. Alice $a=0$ (rows 00 vs 01, $u_0 = (+, +, -, -)$): $n_0(-d/n_0)(+1) + n_1(d/n_1)(+1) = -d + d$ versus $n_0(d/n_0)(+1) + n_1(-d/n_1)(+1) = d - d$. Alice $a=1$ (rows 10 vs 11, $u_1 = (-, +, +, -)$): $n_1(d/n_1)(+1) + n_2(-d/n_2)(+1) = d - d$ versus $n_0(d/n_0)(-1) + n_3(-d/n_3)(-1) = -d + d$. Bob $b=0$ (rows 00 vs 10, $v_0 = (-, +, -, -)$): $n_0(-d/n_0)(-1) + n_1(d/n_1)(+1) = 2d$ versus $n_1(d/n_1)(+1) + n_2(-d/n_2)(-1) = 2d$. Bob $b=1$ (rows 01 vs 11, $v_1 = (+, -, -, -)$): $n_0(d/n_0)(+1) + n_1(-d/n_1)(-1) = 2d$ versus $n_0(d/n_0)(+1) + n_3(-d/n_3)(-1) = 2d$. Each identity is $0 = 0$ or $2d = 2d$, independent of the class counts n_j ; the same closure was verified exactly for all $N = 4, \dots, 64$.
- *Value.* Each row contributes $\mu_{ab} + 2d$, so $S = S_0 + 8d = 2 + 8F^* = 2\sqrt{2}$, with $\text{TV}(p_{ab} \| \rho) = d$ for all ab and NS holding. Hence $F_{\text{NS}}^*(N) \leq F^*$ for all $N \geq 4$.

$N = 3$. In $\mathbb{Q}(\sqrt{2})$, set $d_3 = (\sqrt{2} - 1)/3$; the patterns are $\{M_A, M_B, M_D\}$ on states 1, 2, 3, with response gauge

$$\begin{aligned} u[0, \cdot] &= (-, -, +), & u[1, \cdot] &= (+, -, +), \\ v[0, \cdot] &= (+, -, +), & v[1, \cdot] &= (-, +, +), \end{aligned} \quad (17)$$

and rows (entries are $\frac{1}{3}$ plus d_3 times the coefficient shown)

$$\begin{array}{c|ccc} & \lambda_1 & \lambda_2 & \lambda_3 \\ \hline 00 & -1 & +\frac{1}{2} & +\frac{1}{2} \\ 01 & +\frac{1}{2} & -1 & +\frac{1}{2} \\ 10 & -\frac{1}{2} & +\frac{1}{2} & 0 \\ 11 & +\frac{1}{2} & +\frac{1}{2} & -1 \end{array} \quad (18)$$

Exact $\mathbb{Q}(\sqrt{2})$ arithmetic gives $S = 2 + 6d_3 = 2\sqrt{2}$, per-row TV $= (d_3, d_3, d_3/2, d_3)$, and all four NS residuals exactly zero. Hence $F_{\text{NS}}^*(3) \leq d_3$. (The saved certified array — available from the authors, Appendix C — differs from this printed gauge by the global flip $(u, v) \mapsto (-u, -v)$, which leaves every product q_{ab} — hence S — and the setting-independence of each one-party marginal unchanged.)

Lower and upper bounds together give Eq. (14) for all $N \geq 3$, with explicit algebraic witnesses at every N — an existence-free statement. $\square$

Remark (the whole NS curve on $[0, F^]$.)* Every constraint of the $N \geq 4$ construction is linear in d , so scaling d to any $F \in [0, F^*]$ gives a valid NS model with $\text{TV}(p_{ab} \parallel \rho) = F$ and $S = 2 + 8F$ (feasibility holds throughout). With the global upper bound $S_{\text{NS}} \leq S_{\text{max}} \leq 2 + 8F$, this yields

$$S_{\text{NS}}(N, F) = 2 + 8F \quad \text{on } [0, F^*] \text{ for all } N \geq 4: \quad (19)$$

the NS curve coincides with the signalling-allowed curve up to the quantum point.

D. The algebraic maximum under no-signalling

The silence of Sec. V C extends to the algebraic maximum $S = 4$: the unique NS box that reaches it is the PR box, and the minimum fine-tuning of a local no-signalling decomposition of that box equals the signalling-allowed cap.

Lemma 2 (PR-box uniqueness). *A no-signalling CHSH box with $S = 4$ is uniquely the Popescu-Rohrlich (PR) box [21] (a standard fact, recorded here for completeness): at $(a, b) = (0, 0), (0, 1), (1, 0)$ it outputs $x = y$ with $P(+, +) = P(-, -) = 1/2$, and at $(1, 1)$ it outputs $x = -y$ with $P(+, -) = P(-, +) = 1/2$; all single-party marginals are uniform.*

Proof. $S = E[00] + E[01] + E[10] - E[11] = 4$ with $|E[ab]| \leq 1$ forces $E[ab] = \sigma_{ab}$ at all four corners. Since $x, y \in \{\pm 1\}$, $E[ab] = +1$ means $x = y$ almost surely, and $E[11] = -1$ means $x = -y$ almost surely. Write the NS marginals as $\alpha_a = P(x=+1 | a)$, $\beta_b = P(y=+1 | b)$ (NS makes them independent of the distant setting). The three $x = y$ corners give $\alpha_0 = \beta_0$, $\alpha_0 = \beta_1$, $\alpha_1 = \beta_0$; the $x = -y$ corner gives $\alpha_1 = 1 - \beta_1$. The first three yield a common value r ; the last gives $r = 1 - r$, hence $r = 1/2$, and the box is fully determined. $\square$

Consequence. A fine-tuned local deterministic model that is no-signalling and reaches $S = 4$ is *exactly* a local decomposition of the PR box: $P(x, y | a, b) = \sum_{\lambda} p[ab, \lambda] \mathbf{1}[x = u[a, \lambda]] \mathbf{1}[y = v[b, \lambda]]$ with per-row $\text{TV}(p(\cdot | ab) \parallel \rho) \leq F$. The question is therefore: what is the minimum TV cost of a local decomposition of the PR box, as a function of the number N of hidden states? (The PR box is NS but nonlocal — it has no such decomposition at $F = 0$; the question is how much measurement dependence must be imported to make it local.)

Theorem 8. *For every $N \geq 3$,*

$$F_{\text{NS}}^*(S=4; N) := \min\{F : S_{\text{NS}}(N, F) \geq 4\} = \frac{\lceil N/4 \rceil}{N}. \quad (20)$$

No-signalling neither raises nor lowers the signalling-allowed cap of Theorem 3. The case $N = 2$ is excluded: by Theorem 6, $S_{\text{NS}}(2, F) = 2$ for all F , so $S = 4$ is unattainable there at any fine-tuning level.

Proof. Lower bound. NS restricts the feasible set, so $S_{\text{NS}}(N, F) \leq S_{\text{max}}(N, F)$ and hence $F_{\text{NS}}^*(S=4; N) \geq F(S=4; N) = \lceil N/4 \rceil / N$ by Theorem 3; no NS-specific argument is needed, since the parity-lemma necessity behind that cap ($S = 4 \Rightarrow E[ab] = \sigma_{ab}$ for all ab , so each row must be supported on its correct class, forcing $F \geq n_{ab}/N$ with $\max_{ab} n_{ab} \geq \lceil N/4 \rceil$) does not use NS.

Upper bound — closed-form construction. Use the round-robin patterns of Sec. III C: state i carries misstype $M_{i \bmod 4}$, with class j of size $n_j \in \{\lfloor N/4 \rfloor, \lceil N/4 \rceil\}$. The response gauge is the base realisation per type with one whole class flipped — class D if $N \bmod 4 \in \{0, 1, 2\}$, class C if $N \bmod 4 = 3$ — which leaves every product q_{ab} (hence the patterns and S) unchanged and only fixes the NS marginals. Each row is then supported on its correct class C_{ab} — the class that misses ab carries mass 0 — so $E[ab] = \sigma_{ab}$ and $S = 4$. Writing $a, b, c, d = n_0, n_1, n_2, n_3$, the row masses on the four classes are, in the flip- D case,

$$\begin{array}{c|cccc} & A & B & C & D \\ \hline 00 & 0 & b/N & \frac{1}{2} - b/N & \frac{1}{2} \\ 01 & b/N & 0 & \frac{1}{2} - b/N & \frac{1}{2} \\ 10 & a/N & \frac{1}{2} & 0 & \frac{1}{2} - a/N \\ 11 & a/N & \frac{1}{2} & \frac{1}{2} - a/N & 0 \end{array} \quad (21)$$

(distributed uniformly within each class), and in the flip- C case the same display with b replaced by d and the roles of C and D swapped; all entries are non-negative for every $N \geq 3$. Two facts close the argument. (**NS automatic.**) Substituting the gauge into the four NS residuals, each reduces to the difference of two identical sums (e.g. Alice's $a=0$ residual is $(b/N + (\frac{1}{2} - b/N) - \frac{1}{2}) - (b/N + (\frac{1}{2} - b/N) - \frac{1}{2})$), so all four vanish identically: no NS equation constrains the free entries beyond the row sums. (**TV at or below the cap.**) Expanding the absolute values, each per-row TV is the maximum of two

branch values; in the flip- D case,

$$\begin{aligned} \text{TV}_{00} &= \max(a, N/2 - d)/N, \\ \text{TV}_{01} &= \max(b, N/2 - d)/N, \\ \text{TV}_{10} &= \max(N/2 - b, c)/N, \\ \text{TV}_{11} &= \max(N/2 - b, d)/N. \end{aligned} \quad (22)$$

Since every class count is $\leq \lceil N/4 \rceil$, it remains to check the $N/2 - (\text{class})$ branches: for $N = 4k + r$, $r \in \{0, 1, 2\}$, each is $\leq k + 1 = \lceil N/4 \rceil$ (equality at $r = 0, 2$), and in the flip- C case ($N = 4k + 3$) all four rows land at exactly $\text{TV} = (k + 1)/N = \lceil N/4 \rceil/N$. Hence $\text{TV}(p(\cdot | ab) || \rho) \leq \lceil N/4 \rceil/N$ for all ab and all $N \geq 3$. $\square$

Explicit witnesses (exact; certificate in Appendix C). At $N = 3$ (class sizes 1, 1, 1, 0, flip C), the per-state rows are $p[00, \cdot] = (0, \frac{1}{2}, \frac{1}{2})$, $p[01, \cdot] = (\frac{1}{2}, 0, \frac{1}{2})$, $p[10, \cdot] = p[11, \cdot] = (\frac{1}{2}, \frac{1}{2}, 0)$: $S = 4$, per-row TV $1/3$, all four NS residuals zero, so $F_{\text{NS}}^*(S=4; 3) = \frac{1}{3}$. At $N = 4$ (class sizes 1, 1, 1, 1, flip D), the per-state rows are $p[00, \cdot] = (0, \frac{1}{4}, \frac{1}{4}, \frac{1}{2})$, $p[01, \cdot] = (\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2})$, $p[10, \cdot] = (\frac{1}{4}, \frac{1}{2}, 0, \frac{1}{4})$, $p[11, \cdot] = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4}, 0)$: $S = 4$, per-row TV $1/4$, NS residuals zero — an explicit local decomposition of the PR box at cost $1/4$, so $F_{\text{NS}}^*(S=4; 4) = \frac{1}{4}$. At $N = 5$ (class sizes 2, 1, 1, 1, flip D), the class-mass rows are $p[00, \cdot] = (0, \frac{1}{5}, \frac{3}{10}, \frac{1}{2})$, $p[01, \cdot] = (\frac{1}{5}, 0, \frac{3}{10}, \frac{1}{2})$, $p[10, \cdot] = (\frac{2}{5}, \frac{1}{2}, 0, \frac{1}{10})$, $p[11, \cdot] = (\frac{2}{5}, \frac{1}{2}, \frac{1}{10}, 0)$: $S = 4$, per-row TV $\leq \frac{2}{5}$, NS residuals zero, so $F_{\text{NS}}^*(S=4; 5) = \frac{2}{5}$.

Hiding the fine-tuning from the marginals never raises the price of reaching the algebraic maximum: no-signalling is silent at $S = 4$ as well as at $S = 2\sqrt{2}$, and the cost is the same cap $\lceil N/4 \rceil/N$ as with signalling allowed.

VI. EXPLICIT WITNESS AT THE TSIRELSON POINT

The $N = 4$ optimal point of the signalling-allowed problem, verified by direct arithmetic and as an exact identity in $\mathbb{Q}(\sqrt{2})$ (Appendix C), is: with

$$\begin{aligned} F^* &= \frac{\sqrt{2} - 1}{4} \approx 0.1035534, \\ a &= \frac{1}{4} - F^* \approx 0.1464466, \quad b = \frac{1}{4} + F^* \approx 0.3535534, \end{aligned} \quad (23)$$

the conditional sources $p[ab, \lambda]$ (rows = ab in order 00, 01, 10, 11; columns = $\lambda = 1, \dots, 4$) are

	λ_1	λ_2	λ_3	λ_4
00	$\frac{1}{4}$	a	b	$\frac{1}{4}$
01	a	$\frac{1}{4}$	b	$\frac{1}{4}$
10	b	$\frac{1}{4}$	a	$\frac{1}{4}$
11	b	$\frac{1}{4}$	$\frac{1}{4}$	a

(24)

with deterministic responses

$$\begin{aligned} u[0, \cdot] &= (+, -, -, -), & u[1, \cdot] &= (+, +, +, -), \\ v[0, \cdot] &= (+, +, -, -), & v[1, \cdot] &= (-, -, -, -). \end{aligned} \quad (25)$$

Direct recomputation (independent of any optimisation) gives $S = E[00] + E[01] + E[10] - E[11] = 2\sqrt{2}$ and $F = \max_{ab} \frac{1}{2} \sum_{\lambda} |p[ab, \lambda] - \frac{1}{4}| = (\sqrt{2} - 1)/4$ exactly. A local, deterministic, finite model with a *uniform* source and only $\approx 10.4\%$ settings-conditioned fine-tuning (TV) thus reaches the maximal quantum CHSH violation exactly. (This is the signalling-allowed witness; the no-signalling witness of Theorem 7 has a different gauge and rows, since NS is not gauge-invariant.)

VII. DISCUSSION

A. Interpretation

The fine-tuning cost decreases with N but *saturates* at $N = 4$: adding hidden states beyond four does not lower the minimum fine-tuning needed to reach $2\sqrt{2}$. The bottleneck is the *four CHSH correlators*, not the size of the hidden source — a four-state source already spends its budget optimally across the four (a, b) blocks, and more states just dilute without reducing the per-correlator cost. (The qualitative observation that enlarging the hidden-variable space does not reduce the required relaxation was made earlier for mutual-information-metric singlet models [22]; our saturation is its exact closed-form counterpart in a different metric and target.) By Proposition 1 the floor $F^* \geq (\sqrt{2} - 1)/4 \approx 0.1036$ is model-independent: *any* local CHSH model, on any hidden-variable space, with any source distribution and deterministic or stochastic responses, that reaches the maximal quantum violation must fine-tune at least one conditional source by at least $(\sqrt{2} - 1)/4 \approx 10.4\%$ in TV in its worst settings block.

The no-signalling results sharpen the picture further. The minimum fine-tuning for $S = 2\sqrt{2}$ is achievable *without observable signalling* for every $N \geq 3$ (Theorem 7), and the whole curve up to the quantum point coincides with the signalling-allowed one (Eq. (19) for $N \geq 4$; at $N = 3$ the common curve is $2 + 6F$, by the same scaling argument). The cost of measurement dependence is therefore not paid in observable marginals: a local model can hide its settings-dependence from the single-party statistics and still violate CHSH maximally quantumly, at the same $\approx 10.4\%$ price. This directly addresses the claim of Bacciagaluppi *et al.* that some measurement-independence violations are “signalling in principle” [20]: in this TV metric, the *minimal* fine-tuning for a maximal quantum CHSH violation is NS-silent (for $N \geq 3$). Our statement concerns observable marginals within a fixed model class; it neither refutes nor settles the causal-structure notion of “signalling in principle” of Ref. 20, which is a different question. The two-state case is the converse: under full NS, no fine-tuning at all can produce a Bell violation (Theorem 6) — with two states, hiding the fine-tuning from the marginals makes it useless.

B. One-sided measurement dependence

A natural structural restriction, studied by Hall and Branciard in their mutual-information programme [15], is the *one-sided* causal structure: the hidden source depends on exactly one setting, $p[ab, \lambda] = q[a, \lambda]$ (independent of b), with cost $F_{\text{os}} = \max_a \text{TV}(q(a) \parallel \rho)$; by the $a \leftrightarrow b$ symmetry of CHSH, the Bob-only variant has identical values throughout. Rows then come in pairs, $p[00] = p[01] = q_0$, $p[10] = p[11] = q_1$, and

$$\begin{aligned} S &= \sum_{\lambda} q_0(\lambda) f^A(\lambda) + \sum_{\lambda} q_1(\lambda) f^B(\lambda), \\ f^A(\lambda) &= q_{00}(\lambda) + q_{01}(\lambda), \\ f^B(\lambda) &= q_{10}(\lambda) - q_{11}(\lambda). \end{aligned} \quad (26)$$

each f^A, f^B taking values in $\{-2, 0, +2\}$. The key structural fact: for every even-parity pattern, *exactly one* of $f^A(\lambda), f^B(\lambda)$ is non-zero — each hidden state can actively bend only *one* of the two source rows. That single fact halves the slope of the loophole and removes the saturation phenomenon.

Lemma 3 (one-sided row optimum). *Let $f \in \{-2, 0, +2\}^N$ have k entries equal to $+2$, z entries equal to 0 , and m entries equal to -2 , and set $\mu = 2(k-m)/N$. Then*

$$\begin{aligned} h(f, F) &:= \max\{p \cdot f : p \in \Delta_N, \text{TV}(p \parallel \rho) \leq F\} \\ &\leq \mu + 2F + 2\min(F, m/N) \leq \mu + 4F, \end{aligned} \quad (27)$$

with the exact value a piecewise-linear function of F (water-filling over the three rate classes $+2, 0, -2$) given in Appendix C 1.

Proposition 2. *For every $N \geq 2$,*

$$\begin{aligned} F_{\text{os}}^*(N) &:= \min\{F : S_{\text{os}}(N, F) \geq 2\sqrt{2}\} \\ &= \frac{\sqrt{2}-1}{2} \approx 0.20711, \end{aligned} \quad (28)$$

flat in N from the start, and the one-sided global bound $S_{\text{os}}(N, F) \leq 2 + 4F$ holds for all N and all F .

Proof sketch. Lower bound. Fix the responses and let k_A, m_A (resp. k_B, m_B) count the entries of f^A (resp. f^B) equal to ± 2 . By the key structural fact, $k_A + m_A + k_B + m_B = N$. Lemma 3 bounds each row, and a four-case analysis of whether the two row bounds are above or below 2 — using $\mu_A + \mu_B = 2 - 4(m_A + m_B)/N$, which forces the intercept down whenever both rows are active — gives $S_{\text{os}}(N, F) \leq 2 + 4F$ for all N, F (exact case analysis in Appendix C 1). Hence $S_{\text{os}}(N, F) < 2\sqrt{2}$ for $F < (\sqrt{2}-1)/2$. *Upper bound.* Take $\lceil N/2 \rceil$ states of a pattern with $f^A = +2$ and $\lfloor N/2 \rfloor$ states of one with $f^B = +2$; then row A has $(k, m, z) = (\lceil N/2 \rceil, 0, \lfloor N/2 \rfloor)$ and row B the mirror, both in the slope-2 regime of Lemma 3, so $S_{\text{os}}(N, F) \geq 2 + 4\min(F, \lfloor N/2 \rfloor/N)$. Since $(\sqrt{2}-$

$1)/2 < \lfloor N/2 \rfloor/N$ for all $N \geq 2$ (an exact $\mathbb{Q}(\sqrt{2})$ identity, checked in the certificate), the construction reaches $S = 2 + 4F_{\text{os}} = 2\sqrt{2}$ at $F_{\text{os}} = (\sqrt{2}-1)/2$. $\square$

Consequences. The one-sided cost equals the full-structure value at $N = 2$ exactly: adding hidden states never helps, so in cost terms the one-sided structure is worth throwing away every state beyond two. The penalty factor relative to the full causal structure is *exactly* 2 for $N \geq 4$ (and $3/2$ at $N = 3$), since $F_{\text{os}}^*(N)/F^*(N) = \min(N, 4)/2$. The cap to $S = 4$ under one-sided dependence is $\lceil N/4 \rceil/N$ (proved): exactly twice the full-model cap $\lceil N/4 \rceil/N$ for $N \equiv 0, 3 \pmod{4}$, and a factor $\geq \frac{3}{2}$ (tending to 2) for $N \equiv 1, 2 \pmod{4}$ with $N \geq 5$ (at $N = 2$ the two caps coincide, both equal to $\frac{1}{2}$). Hall and Branciard’s mutual-information metric gives the same qualitative ordering — one-sided ≈ 0.128 bits versus causal ≈ 0.080 bits, a factor ≈ 1.6 — whereas the TV metric gives the factor exactly 2; the two metrics are incommensurable, so no cross-metric ordering of the absolute numbers follows (Sec. VII E). The exact one-sided curves $S_{\text{os}, \text{max}}(N, F)$ for $N = 2, \dots, 8$ are certified in Appendix C 1: for even N the curve is the single segment $\min(2 + 4F, 4)$ (proved for *all* even N), and for odd N it bends once at $F = \lfloor N/2 \rfloor/N$ (proved for $N = 3, 5, 7$; the bend line for odd $N \geq 9$ is conjectured).

C. Relation to prior work

Table I positions this work relative to the closest prior literature (all entries independently verified against arXiv; a completed 2015–2026 sweep with full search log is described in Appendix D).

Three similar numbers — disambiguation. (i) Hall’s L^1 convention: $M_{\text{singlet}} = \frac{2}{3}(\sqrt{2}-1) \approx \mathbf{0.276}$. (ii) Fraction-of-independence-lost convention ($M/2$): $(\sqrt{2}-1)/3 \approx \mathbf{0.1381}$ — Hall’s $S = 2\sqrt{2}$ cost in that convention, and the $\eta = 1$ floor of Ref. 17. (iii) Our per-block TV metric: $F^* = (\sqrt{2}-1)/4 \approx \mathbf{0.1036}$. These come from different metrics on different model classes; no ordering of the numbers follows, and none contradicts any other. Hall’s metric is pairwise (no fixed baseline) and his bound is derived for no-signalling models; ours measures each conditional source $p(\cdot \mid ab)$ against a fixed uniform baseline and allows (optionally forbids) signalling. Our TV is a different, non-information-theoretic measure of the same statistical-independence violation; explicit conversion bounds between the conventions for the *same* model, and the identification of our model class with the general (retrocausal) causal structure of Ref. 15, are worked out in Sec. VII E.

Metric dependence of the answer. How much measurement dependence is “enough” is known to depend strongly on both the metric and the target correlations. In the polytope parametrisation of Pütz *et al.*, any nonzero residual measurement independence can be detected by suitable quantum correlations [12, 13]; in

TABLE I. Comparison of fine-tuning metrics and values at the CHSH Tsirelson point $S = 2\sqrt{2}$. “Ret.” = fraction of independence retained.

Work	Metric	Value at $S = 2\sqrt{2}$
Hall [9]	pairwise L^1 distance $M \in [0, 2]$ between setting-pair sources; NS, continuous λ	$M_{\text{singlet}} = \frac{2}{3}(\sqrt{2} - 1) \approx 0.276$ (ret. ≈ 0.862)
Hall & Branciard [15]	mutual information $I(X, Y : \Lambda)$ in bits, fixed causal structures	causal ≈ 0.080 ; retrocausal ≈ 0.046 ; one-sided ≈ 0.128 ; superdet $= 2$ bits
Friedman <i>et al.</i> [14]	per-observer Hall-type parameters, combined relaxations	tight general bounds (no single number)
Takakura <i>et al.</i> [16]	Hall’s M + hidden information $H = 1 - \max_{\lambda} p(\lambda)$	$S \leq 2 + \min\{3M, \frac{3}{4}M + 2H, 2\}$, necessary & sufficient
Pal <i>et al.</i> [19]	ensemble-divergence δ_{ens} (TV-type) for drifting preparations; MI preserved	$ S \leq 2 + 6\delta_{\text{ens}}$ (not claimed tight)
Alai [17]	Hall-type variational M (normalised as Hall’s $M/2$) + detector efficiency η_{eff}	closed form $M(S, \eta)$; at $\eta = 1$: $(\sqrt{2}-1)/3 \approx 0.1381$
Alai [18]	Hall’s M , $F := M/2$ (GHZ–Mermin)	$F(n) = R/[2(R+1)]$ ($R = 2^{\lfloor (n-1)/2 \rfloor}$) for $n = 3, \dots, 13$; CHSH consistency value $(\sqrt{2}-1)/3$
This work	per-setting TV($p(\cdot ab)\ \rho$) from a fixed baseline source	$F^* = (\sqrt{2} - 1)/\min(N, 4) \approx 0.1036$ ($N \geq 4$), exact; NS-silent for $N \geq 3$

Hall’s M [9], in mutual information [10, 15], and in our per-setting TV metric, a modest and quantitatively fixed amount reproduces the CHSH Tsirelson value. Our result should be read as one exactly solved point in that metric-dependent landscape, not as a metric-independent price of the loophole.

Novelty, stated as a search result. Within arXiv-indexed literature through 2026-09-02, no prior use was found of (i) the per-setting TV-from-a-fixed-source metric, (ii) the exact closed forms $F^* = (\sqrt{2} - 1)/\min(N, 4)$ with $N = 4$ saturation together with the exact curves and

cap $\lceil N/4 \rceil/N$, or (iii) an NS-silence result of this kind. This is stated as a search result, not “proven novel”: coverage was arXiv-centric (~ 16 papers read at full or abstract depth, the rest at catalogue level); paywalled journal content not mirrored on arXiv, non-arXiv repositories, and books/theses are not covered. The closest prior work is Ref. 17 — exact minimum measurement dependence for CHSH via LP (machine precision on a 32-point grid, with an exact $\mathbb{Q}(\sqrt{2})$ certificate at one point), in a Hall-type metric normalised as Hall’s $M/2$, plus detector-efficiency degrees of freedom; its $\eta = 1$ floor at $2\sqrt{2}$, $(\sqrt{2} - 1)/3 \approx 0.1381$, is a different number from our $F^* = (\sqrt{2} - 1)/4 \approx 0.1036$ because the metrics differ — no contradiction. Together with Refs. 16, 18, and 19, this shows that “exact minimum measurement dependence” is an active 2026 front; our contribution is a specific metric with clean closed forms, plus the NS sub-case, which none of them covers.

D. Implications for proposed superdeterministic models

The model-independence of Proposition 1 means that the floor $F^* \geq (\sqrt{2} - 1)/4 \approx 10.4\%$ is a price attached to the *statistics* of any local account of the Tsirelson value, independent of the causal story told for the settings-dependence. Several explicit models have been proposed in the superdeterminism literature, and all fall on the priced side of this bound. Brans’s fully causal model [7] — in which the hidden variable determines both outcomes and settings and reproduces the singlet correlations for arbitrary spin-1/2 directions, as described in Ref. 9 — is a direct instance of our model class with a settings-dependent conditional source $p(\lambda \mid a, b)$; if it reaches $S = 2\sqrt{2}$, some settings block must carry $\geq 10.4\%$ TV. ’t Hooft’s program of classical deterministic underpinnings [23] claims that any quantum model can be mimicked to arbitrary finite accuracy by classical deterministic dynamics, with Bell’s theorem inapplicable because the realistic state distribution depends on the measurement basis; reaching $S = 2\sqrt{2} - \varepsilon$ in such a construction forces $F \geq (\sqrt{2} - 1)/4 - O(\varepsilon)$. In Palmer’s discretised Hilbert-space models [24, 25] the hidden variable determines the actual apparatus settings, so conditioning on observed settings restricts λ to a proper subset of its domain; since total variation is non-increasing under coarse-graining, any level of description that locally reproduces $2\sqrt{2}$ forces $F \geq 10.4\%$ at the fine-grained level as well — “invisible to coarse-graining” arguments cannot lower the cost below the floor. (Palmer’s separate claim that his framework is not fine-tuned with respect to a p -adic metric on its state space concerns a different metric and is not addressed by our bound.) The most explicit recent construction we examined, Garza and Hance’s conservation-law model [26], keeps a fixed uniform baseline source while measurement error removes a settings-dependent band from the hidden-variable do-

main; at its best-fit parameter it reaches $|S| \approx 2.826$, within 0.08% of the Tsirelson value, with worst-block TV of $\approx 17\text{--}29\%$ — a factor $\approx 1.7\text{--}2.8$ above the floor — and the inequality $S \leq 2 + 8F$ holds across its parameter space (verified numerically; Appendix C). [27] We also note a point of self-consistency in that construction: their correlator (their Eq. (14)) satisfies $|E(\theta; \Delta L)| \leq 1$ for all θ if and only if $\Delta L \geq \frac{1}{2}$, and their model already takes $\Delta L \geq 0.5$ — nothing they claim lies in the $|E| > 1$ regime. We stress what the bound does and does not say: it is compatible with every one of these models, and it rules out none of them; it fixes a quantitative price floor for the loophole, so that “a little fine-tuning” in the sense of $< 10.4\%$ TV per settings block is insufficient, by theorem, for any local reproduction of the maximal quantum violation. Whether such a deviation is physically plausible is a separate question. Experimental model selection weighs against superdeterministic causal structures [28], and Vieira, Ramanathan and Cabello have shown theoretically that local models with *partially* relaxed measurement independence (and parameter independence) can be excluded by Bell-like experiments with many settings on high-dimensional entangled states [29]; that exclusion does not cover the two-setting CHSH scenario priced here, but it delimits how far the loophole can be pushed.

E. Metric comparison

The fine-tuning cost can be measured in four conventions (Table II). They all measure the same statistical-independence violation and differ only in (i) whether a fixed baseline ρ is used and (ii) the L^1 -vs-half- L^1 -vs-entropy functional — which is exactly why explicit conversion bounds exist between them.

TABLE II. The fine-tuning conventions of this work and the closest literature. $p_{ab} = p(\cdot \mid a, b)$, $\text{TV}(p \parallel q) = \frac{1}{2} \|p - q\|_1$, $p_{\text{avg}} = \frac{1}{4} \sum_{ab} p_{ab}$.

Metric	Definition
F (this work)	$\max_{ab} \text{TV}(p_{ab} \parallel \rho)$: worst case from a <i>fixed</i> baseline ρ
M_{model}	$\max_{ab, a'b'} \text{TV}(p_{ab} \parallel p_{a'b'})$: pairwise, no baseline
M (Hall [9])	$\max_{ab, a'b'} \ p_{ab} - p_{a'b'}\ _1 = 2M_{\text{model}}$
$I(X, Y; \Lambda)$ (H&B [15])	$H(\Lambda) - \frac{1}{4} \sum_{ab} H(\Lambda \mid a, b)$, in bits (equal to their $I(X, Y; \Lambda)$ for uniformly distributed settings, as they assume)

Conversion bounds. For every local CHSH model and every baseline ρ :

1. $M_{\text{model}} \leq 2F$, by the triangle inequality, $\text{TV}(p_{ab} \parallel p_{a'b'}) \leq \text{TV}(p_{ab} \parallel \rho) + \text{TV}(p_{a'b'} \parallel \rho) \leq 2F$;

2. $F \leq M_{\text{model}} + \text{TV}(p_{\text{avg}} \parallel \rho)$, by $\text{TV}(p_{ab} \parallel \rho) \leq \text{TV}(p_{ab} \parallel p_{\text{avg}}) + \text{TV}(p_{\text{avg}} \parallel \rho)$ and taking the maximum over ab ;
3. no bound $F \leq f(M_{\text{model}})$ holds for a *fixed* external ρ : if all four rows equal some $q \neq \rho$, then $M_{\text{model}} = 0$ while $F = \text{TV}(q \parallel \rho) > 0$.

Bound (i) is tight: for both $N = 4$ witnesses (Secs. V and VI) the six pairwise TVs take values in $\{\delta, 2\delta\}$ with $\delta = F^*$, so $M_{\text{model}} = 2F^*$ exactly — but it is not saturated by every F^* -optimal witness: the $N = 3$ NS witness of Theorem 7 has $M_{\text{model}} = \frac{3}{2}F^*$ (slack $F^*/2$).

Our witnesses in the other conventions. The F^* -optimal $N = 4$ witnesses cost $M_{\text{model}} = 2F^* = (\sqrt{2} - 1)/2 \approx 0.207$ in the pairwise-TV convention, i.e. Hall’s full- L^1 value $M = \sqrt{2} - 1 \approx 0.414$ — which in his “fraction of independence retained” reads $1 - M/2 = (3 - \sqrt{2})/2 \approx 0.793$ — and they carry $I(X, Y; \Lambda) \approx 0.056$ bits; the $N = 3$ NS witness carries ≈ 0.055 bits.

Different metrics select different optima. Our TV-optimal witness is not M -optimal: Hall’s bound $S \leq \min\{2 + 3M, 4\}$ — stated by Hall for deterministic no-signalling models; we re-prove it in Appendix C using only local determinism — at $M = \sqrt{2} - 1$ allows up to $S = 3\sqrt{2} - 1 \approx 3.24$, whereas our witness sits at $2\sqrt{2} \approx 2.83$; nor is it MI-optimal, its ≈ 0.056 bits exceeding the relevant floor of ≈ 0.046 bits. Different metrics select different optimal models; this is why the three numbers of the disambiguation above (0.276/0.138/0.104) coexist without contradiction.

Model-class identification. Ref. 15 minimizes $I(X, Y; \Lambda)$ under fixed causal graphs. Its causal structure imposes the additional factorization constraint that the two *settings* be conditionally independent given the hidden variable, $p(a, b \mid \lambda) = p(a \mid \lambda)p(b \mid \lambda)$ (written $p(x, y \mid \lambda) = p(x \mid \lambda)p(y \mid \lambda)$ in Ref. 15, whose x, y denote settings), which our closed-form witnesses do *not* satisfy (verified exactly in $\mathbb{Q}(\sqrt{2})$ for every witness of this paper); equivalently, since Ref. 15 proves its causal floor ≈ 0.080 bits is the minimum over that class, no causal model can reach $2\sqrt{2}$ at ≈ 0.056 bits. Our model class — free $p(\lambda \mid a, b)$ against a fixed ρ — is therefore H&B’s *general* (retrocausal) structure, for which the relevant floor at $S = 2\sqrt{2}$ is $I_R \approx 0.046$ bits, not the causal floor $I_C \approx 0.080$; our witnesses’ ≈ 0.056 bits sit above it, as they must.

A sharper bound in our variable. Hall’s $S \leq 2 + 3M$ becomes $S \leq 2 + 12F$ when written in our variable ($M = 2M_{\text{model}} \leq 4F$), versus our direct $S \leq 2 + 8F$: TV from a fixed ρ yields the sharper CHSH bound because all four rows share one center.

Proposition 3 (aggregation robustness). *(i) For every local CHSH model — any source, deterministic or stochastic responses — $S \leq 2 + 8F_{\text{avg}}$, where $F_{\text{avg}} = \frac{1}{4} \sum_{ab} \text{TV}(p(\cdot \mid ab) \parallel \rho)$. (ii) The $N = 4$ witness of Sec. VI has all four per-row TVs equal to $d = (\sqrt{2} - 1)/4$, so*

$$F_{\text{avg}}^*(4) := \min\{F_{\text{avg}} : S = 2\sqrt{2} \text{ achievable}\} = d = F^*(4).$$

Proof sketch. (i) The per-row estimate $|\sigma_{ab}E[ab] - \mu_{ab}| \leq 2\text{TV}(p(\cdot | ab) || \rho)$ behind Proposition 1 uses only the *row's own* TV; summing the four rows with row-specific TVs gives $S \leq S_0 + 2 \sum_{ab} \text{TV}(p(\cdot | ab) || \rho) = S_0 + 8F_{\text{avg}} \leq 2 + 8F_{\text{avg}}$. (ii) An exact $\mathbb{Q}(\sqrt{2})$ certificate (Appendix C) re-derives the witness arithmetic: the four per-row TVs are all exactly d , so its average-TV cost is d , and (i) gives the reverse inequality. $\square$

The headline is therefore insensitive to the aggregation choice at $N = 4$: fine-tuning cannot be made cheaper by spreading it across the setting blocks rather than concentrating it in one. What is *not* claimed: $F_{\text{avg}}^*(N)$ for $N > 4$ has not been computed; only the model-independent floor $(\sqrt{2} - 1)/4$ and the $N = 4$ equality are proved.

F. Scope, caveats, and verification

The exact *curves* are certified for $N = 2, \dots, 14$ (exhaustive enumeration; Theorem 5 prints $N \leq 8$); for $N \geq 15$ the first segment, the cap, and the headline F^* are proved for all N , while the full bend region is left to numerics. The exact finite- N statements are certified for the *uniform* source; Proposition 1 shows the lower bound $F^* \geq (\sqrt{2} - 1)/4$ does not depend on the source distribution or on determinism, so no extension can lower the floor. Finally, F^* is a mathematical minimum within the model class — it is not compared to any physical prior on how much fine-tuning a real source could have.

Every result in this paper is checkable without trusting the prose: the exact statements carry re-runnable exact-arithmetic certificates (rational and $\mathbb{Q}(\sqrt{2})$, no floating point; exit code 0 iff every assertion passes), and an independent LP pipeline reproduces the same values to $\leq 1.8 \times 10^{-15}$. The full verification record, including how to re-run each check, is in Appendix C.

VIII. CONCLUSION

We quantified the price of the measurement-dependence escape route for Bell/CHSH in a per-setting total-variation metric. The price has a closed form: $F^* = (\sqrt{2} - 1)/\min(N, 4)$, saturating at four hidden states; the global bound $S \leq 2 + 8F$ is model-independent and tight; the cap to the algebraic maximum is $\lceil N/4 \rceil / N$; the no-signalling constraint is exactly silent about the minimum for all $N \geq 3$ — at both the Tsirelson value and the algebraic maximum, where the unique NS box is the PR box — with explicit algebraic witnesses, while at $N = 2$ it forbids any violation; and the one-sided causal structure pays exactly twice the full-structure cost, flat in N . The superdeterminism loophole, in this metric, costs a concrete $\approx 10.4\%$ of TV per settings block — and that

cost can be hidden from the observed marginals. The systematic comparison with Hall's M and the conventions of Refs. 15 and 17 is given in Sec. VII E, and the generalisation of the program beyond CHSH — CGLMP, Mermin, and the Peres–Mermin square — is carried out in the companion paper [30], which identifies the structural conditions (parity frustration, the number K of extremal conditions, the minimal miss set) under which the saturation form survives. Natural next steps are extending the exact-curve certificates to $N \geq 15$ and to the open I3322 inequality.

NOTE ON PROVENANCE AND CORRECTIONS

This paper originated as an experiment in machine-generated research. A locally hosted open-weight language model, Qwen3.8-27B, was asked to identify a scientific problem of interest, research the literature, carry out the mathematics and computations, and write a publication. The model was shepherd through the process by the human author, who acted as project manager and is listed as the author; but the scientific and technical content — the choice of problem and metric, the theorems and proofs, the certificates, and the text — is the product of the model, with minimal scientific or technical input from the human author.

On 2026-09-03 the manuscript, its Supplemental Material (now Appendices A–D), the companion paper and the verification pipeline were critically reviewed by a second model, Claude Fable 5.1 (Anthropic). The reviewer re-derived the results by hand and by independently written exact-arithmetic and linear-programming checks (sharing no code with the project's certificates), re-ran every certificate, and audited the bibliography and every literature claim against primary sources. Every theorem, curve, witness and certificate tested was confirmed; no result changed. The following corrections were then made by the reviewer:

1. A duplicated equation label made the proof of Theorem 7 cite the wrong equation; fixed.
2. Bibliography: the Tsirelson reference had the wrong volume, pages and year (now Cirel'son, Lett. Math. Phys. **4**, 93 (1980)); the Shimony reference was garbled (its title, editors, publisher and pages did not match the source it claimed to have been verified against; now the Kamefuchi volume, pp. 225–230); the DOI of Ref. 29 had transposed digits; Jarrett's page range was wrong; the Popescu–Rohrlich reference was added. The central prior works of Barrett and Gisin [10], Pütz *et al.* [12, 13] and Koh *et al.* [11], which had been omitted, were added to the Introduction and to Sec. VII C.
3. Sec. VII E: the Hall–Branciard factorization constraint had been transcribed in their notation $(x, y$

= settings), which in this paper’s notation (x, y = outcomes) made the sentence false; rewritten in terms of the settings.

4. Sec. VII B: the one-sided/full cap ratio was stated as $\geq 3/2$ for $N \equiv 1, 2 \pmod{4}$, which fails at $N = 2$; restricted to $N \geq 5$.
5. Table I and Sec. VII C: Alai’s measure is normalised as Hall’s $M/2$, which table and text stated inconsistently, and the description of his certificates was overstated; both corrected.
6. The literature-sweep cutoff was advanced to 2026-09-02, matching the project’s delta check, and the existing exact-curve certificate for $N = 9, \dots, 14$, which had not been cited, is now cited.
7. Clarifications: the uniform-baseline origin of the non-monotone cap (Sec. IV B); the standard status of the PR-box lemma; the scope of Hall’s bound and of the Hall–Branciard mutual-information formula; Palmer’s p -adic argument; the role of Ref. 29; the coincidence of the $N = 4$ and $N = 8$ curves in Fig. 1; and the cross-references to the appendices.
8. The abstract was rewritten to state the contribution and its scope first.
9. For arXiv posting the Supplemental Material was merged into this paper as Appendices A–D; within it the blanket “no floating point” statement was qualified, the $N \geq 9$ curve statement now cites the existing `curves_highN` certificate (independently re-derived), the sweep cutoff date was updated, and the precision of the mutual-information assertions is stated.

The companion paper received analogous corrections, the most important being that its CGLMP target was the maximally-entangled-state value rather than the larger known quantum value of Acín *et al.*; see the corresponding note there. The full review record is kept with the project files. The final form should therefore be read as a combined work: primarily produced by the Qwen model, critically reviewed and corrected by Claude Fable 5.1, and directed by the human author.

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These appendices contain (A) the table of even-parity product patterns, (B) details of the exact-curve certificate, (C) the complete verification record with re-run instructions, and (D) the literature sweep underlying

the novelty statements. The exact certificates are self-contained Python scripts using exact rational or $\mathbb{Q}(\sqrt{2})$ arithmetic; the floating-point parts (LP cross-checks, mutual-information numerics, the Garza–Hance evaluation) are labelled as such. Each script exits with code 0 if and only if every assertion passes.

Appendix A: Product patterns

The eight even-parity patterns $q = (q_{00}, q_{01}, q_{10}, q_{11})$ with $q_{00}q_{01}q_{10}q_{11} = +1$, their CHSH values $C(q) = q_{00} + q_{01} + q_{10} - q_{11}$, and their relation to the sign vector $\sigma = (+, +, +, -)$:

Pattern	$C(q)$	#diff. from σ	Name	Misses
$(-1, +1, +1, -1)$	+2	1	M_A	$q_{00} = +1$
$(+1, -1, +1, -1)$	+2	1	M_B	$q_{01} = +1$
$(+1, +1, -1, -1)$	+2	1	M_C	$q_{10} = +1$
$(+1, +1, +1, +1)$	+2	1	M_D	$q_{11} = -1$
$(+1, -1, -1, +1)$	-2	3	—	(negative of M_A)
$(-1, +1, -1, +1)$	-2	3	—	(negative of M_B)
$(-1, -1, +1, +1)$	-2	3	—	(negative of M_C)
$(-1, -1, -1, -1)$	-2	3	—	(negative of M_D)

TABLE III. The eight even-parity product patterns. Each miss-type pattern differs from σ at exactly one coordinate; the parity lemma follows because σ itself has product -1 .

Appendix B: Exact-curve certificate

Theorem 5 (exact curves for $N = 2, \dots, 8$) is certified by exhaustive evaluation of the finite-maximum formula

$$S_{\max}(N, F) = \max_Q [S_0(Q) + \sum_{ab} g_{ab}(F)], \quad (\text{B1})$$

over all $\binom{8+N-1}{N}$ multisets Q of even-parity patterns, in exact rational arithmetic:

N	$\binom{8+N-1}{N}$	bend point(s)	$S_{\max}$ at bends	cap $F(S=4; N)$
2	36	—	—	$\frac{1}{2}$
3	120	—	—	$\frac{1}{3}$
4	330	—	—	$\frac{1}{4}$
5	792	$F = \frac{1}{5}$	$\frac{18}{5}$	$\frac{3}{5}$
6	1716	$F = \frac{1}{6}$	$\frac{10}{3}$	$\frac{1}{2}$
7	3432	$F = \frac{1}{7}$	$\frac{22}{7}$	$\frac{3}{7}$
8	6435	—	—	$\frac{1}{4}$

TABLE IV. Enumeration sizes and certified bend values. The upper envelope of the exact affine functions agrees with the round-robin lower bound on every cell; slopes are in $\{0, 2, 4, 6, 8\}$ as the formula forces.

For $N = 9, \dots, 14$ the same exhaustive certificate (`curves_highN.py` $\rightarrow$ `curves_highN.out`; $\binom{N+7}{7}$ multisets, 11 440 at $N = 9$; 32 s in total) confirms $S_{\max}(N, F) = \min\{2 + \sum_{ab} 2 \min(F, n_{ab}(Q_N)/N), 4\}$

on every cell, with the round-robin counts $n_{ab}(Q_N) \in \{\lfloor N/4 \rfloor, \lceil N/4 \rceil\}$; this was re-derived by an independently written exhaustive enumeration in the review of 2026-09-03 (`curves_N9to14.py`, grid points and cell midpoints, exact rationals). For $N \geq 15$ only the first segment $S_{\max} = 2 + 8F$ on $[0, 1/N]$, the cap $\lceil N/4 \rceil/N$, and the headline value $S_{\max}(N, F^*) = 2\sqrt{2}$ are proved; the bend region is left to numerics.

Appendix C: Verification record and code availability

Every result in this paper is checkable without trusting the prose. Three independent routes are provided.

1. One-sided: exact row formula and case analysis

The one-sided row optimum of Lemma 3 has the exact piecewise-linear form

$$\begin{aligned} h(f, F) &= \mu + G^*(F), \quad G^*(F) = \max_{\delta \in [0, F]} [P(\delta) + N(\delta)], \\ P(\delta) &= 2 \min(\delta, k(N-1)/N) \\ &\quad - 2 \max(0, \delta - k(N-1)/N - z(N-1)/N), \\ N(\delta) &= 2 \min(\delta, m/N) \\ &\quad - 2 \max(0, \delta - m/N - z/N), \end{aligned} \quad (C1)$$

where $f \in \{-2, 0, +2\}^N$ has k entries $+2$, z entries 0 , m entries -2 , and $\mu = 2(k-m)/N$. The form is water-filling: with $d = p - \rho$ and $\delta = \sum_{d > 0} d_\lambda \leq F$, the positive part of d (per-entry cap $1 - 1/N$) is placed at rate $+2$ on the k entries and then at rate 0 on the z entries, giving $P(\delta)$; the negative part (per-entry cap $1/N$) is placed at rate $+2$ on the m entries (mass removed from a -2 entry gains $+2$) and then at rate 0 , giving $N(\delta)$; overlapping placements cancel L^1 norm, never gain. The script cross-checks Eq. (C1) against an independent LP for all (k, z, m) types, $N = 2, \dots, 8$, on a dense F -grid including every breakpoint (4109 pairs, maximum deviation 1.1×10^{-15}).

The bound $S_{\text{os}}(N, F) \leq 2 + 4F$ of Proposition 2 follows by applying the lemma to the two rows. Write $Z_A := \mu_A + 2F + 2 \min(F, m_A/N)$ and Z_B likewise, where μ_A, μ_B and m_A, m_B refer to f^A, f^B ; then $X := h(f^A, F) = \min(2, Z_A)$, $Y = h(f^B, F) = \min(2, Z_B)$, $S \leq X + Y$, and by the key structural fact $k_A + m_A + k_B + m_B = N$ with $\mu_A + \mu_B = 2 - 4(m_A + m_B)/N$. *Case (a)* $Z_A, Z_B \leq 2$: $S \leq Z_A + Z_B = 2 - 4(m_A + m_B)/N + 4F + 2 \min(F, m_A/N) + 2 \min(F, m_B/N) \leq 2 + 4F$. *Case (b)* $Z_A > 2 \geq Z_B$: $X = 2$ and one needs $Z_B \leq 4F$, i.e. $(k_B - m_B)/N + \min(F, m_B/N) \leq F$. From $Z_A > 2$: $(k_A - m_A)/N + F + \min(F, m_A/N) > 1$; since $-m_A/N + \min(F, m_A/N) \leq 0$, this gives $k_A > N(1 - F)$, hence $k_B = N - k_A - m_A - m_B < NF - m_A - m_B$ and $(k_B - m_B)/N + \min(F, m_B/N) < F - (m_A + 2m_B)/N +$

$\min(F, m_B/N) \leq F$, so $S \leq 2 + Z_B \leq 2 + 4F$. *Case (c)* is symmetric to (b). *Case (d)* $Z_A, Z_B > 2$: adding the two strict inequalities gives $(k_A + k_B - m_A - m_B)/N + 2F + \min(F, m_A/N) + \min(F, m_B/N) > 2$, but the left side is $\leq 1 - (m_A + m_B)/N + 2F < 2$ for $F < \frac{1}{2}$ (using $k_A + k_B \leq N - (m_A + m_B)$ and $\min(F, m_A/N) \leq m_A/N$); for $F \geq \frac{1}{2}$ one has $X = Y = 2$ and the trivial bound $S \leq 4 \leq 2 + 4F$ suffices.

The status of the exact one-sided curves certified by `onesided.py` is: for even N , the single-segment form $S_{\text{os}, \max}(N, F) = \min(2 + 4F, 4)$ is proved for *all* even N (upper: the case analysis plus $S \leq 4$; lower: the Q_N^* construction); for odd $N = 2M + 1$, the two-segment form $2 + 4F$ on $[0, M/N]$, $(6M + 2)/(2M + 1) + 2F$ on $[M/N, (M + 1)/N]$, then 4 , is certified by exact enumeration for $N = 3, 5, 7$ and is conjectured for odd $N \geq 9$; the cap $F_{\text{os}}(S=4; N) = \lceil N/2 \rceil/N$ is proved for all N : $S = 4$ forces each row onto its $f^A = +2$ (resp. $f^B = +2$) support, costing $\text{TV} \geq (N - k_A)/N$ and $(N - k_B)/N$ with $k_A + k_B \leq N$, and the round-robin split attains the maximum.

2. Exact certificates (no floating point)

- `audit1.py` $\rightarrow$ `audit1.out` — rational arithmetic. **Part A:** the exact $S_{\max}(N, F)$ curves and cap minimum over *all* multisets for $N = 2, \dots, 8$ (Theorems 3 and 5; Theorem 4 follows from Theorem 2 for $N \geq 4$ and from Theorem 5 for $N = 2, 3$, and is float-cross-checked in Part B). **Part B:** floating-point cross-check of F^* at $N = 2, \dots, 6$: $S_{\max}(N, F_N^*) = 2.828427124746$ to 12 digits (margins to the runner-up construction: 0 for $N = 2, 3, 5, 6$ — multiple optimal Q 's; 2.07×10^{-1} for $N = 4$, where Q_4 is the unique optimiser). **Part C:** residuals of the saved witness arrays, attributed to the bisection tolerance (10^{-5}) of the search pipeline. **Part D:** the $N = 4$ witness table of Sec. VI as an *exact identity* in $\mathbb{Q}(\sqrt{2})$: $E[ab] = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2})$, $S = 2\sqrt{2}$, per-row $\text{TV} = F^* = (\sqrt{2} - 1)/4$ exactly. **Part E:** the two-state NS certificate (Theorem 6): all 256 response vertices, union-find closure over the four NS equations, 1312 candidate affine functions, every one ≤ 2 on $F \in [0, \frac{1}{2}]$.
- `ns_general.py` $\rightarrow$ `ns_general.out` — Theorem 7 (NS headline) certificates. **Part A:** the $N \geq 4$ closed form verified exactly — row constraints and all four NS identities with d formal, for $N = 4, \dots, 64$; plus the feasibility inequality $\lfloor N/4 \rfloor \geq (\sqrt{2} - 1)N/4$ for $N = 4, \dots, 999$. **Part B:** the $N = 3$ witness in $\mathbb{Q}(\sqrt{2})$: $S = 2\sqrt{2}$ exactly, per-row $\text{TV} (d_3, d_3, d_3/2, d_3)$, all four NS residuals 0 . **Part C:** the closed forms versus the saved witness arrays (maximum deviation 5.6×10^{-17}).
- `prbox_ns.py` $\rightarrow$ `prbox_ns.out` — Theorem 8 (NS algebraic maximum) certificates; 28 checks, exit code 0 . **Part A:** exact rational recomputation of the signalling

cap $\min_Q \max_{ab} n_{ab}(Q) = \lceil N/4 \rceil$ over *all* even-parity multisets for $N = 2, \dots, 8$ (self-contained lower bound, with the monotonicity $S_{NS} \leq S_{\max}$). **Part B:** the closed-form construction of Sec. VD rebuilt from the formula in exact rational arithmetic for $N = 3, \dots, 300$ plus spot checks at $N = 1000, 4096, 10000$: row sums 1, non-negativity, $E[ab] = \sigma_{ab}$ (hence $S = 4$), per-row $TV \leq \lceil N/4 \rceil / N$, all four NS residuals 0; explicit witnesses printed for $N = 3, 4, 5$ (the $N = 4$ witness is an explicit local decomposition of the PR box at cost $1/4$). **Part C:** independent HiGHS LP (float) feasibility at $F = \lceil N/4 \rceil / N$ for $N = 3, \dots, 16$ — a search cross-check only; the claims rest on Parts A–B.

- **onesided.py** $\rightarrow$ **onesided.out** — Proposition 2 (one-sided measurement dependence) certificates. **Part 0:** the exact row formula Eq. (C1) cross-checked against an independent LP for all (k, z, m) types, $N = 2, \dots, 8$ (4109 pairs, maximum deviation 1.1×10^{-15}). **Part A:** the exact $S_{os, \max}(N, F)$ curves for $N = 2, \dots, 8$ (exact upper hull over all $\binom{8+N-1}{N}$ multisets), asserting the closed form of Sec. C1, including $S_{os, \max}(N, 0) = 2$ and first-segment slope 4. **Part B:** the headline $F_{os}^*(N) = (\sqrt{2} - 1)/2$ as an exact $\mathbb{Q}(\sqrt{2})$ identity, including $(\sqrt{2} - 1)/2 < \lfloor N/2 \rfloor / N$ for $N = 2, \dots, 8$, and the cap $\lceil N/2 \rceil / N$. **Parts C1–C3:** the one-sided-NS variants (not cited elsewhere in this paper): $S_{NS, os}(2, F) \equiv 2$ and $S_{NS, os}(3, F) \equiv 2$ at all F , and $F_{NS, os}^*(N) = (\sqrt{2} - 1)/2$ for $N \geq 4$ with explicit $\mathbb{Q}(\sqrt{2})$ witnesses for $N = 4, \dots, 8$.
- **metric_compare.py** $\rightarrow$ **metric_compare.out** — the metric comparison of Sec. VII E. Exact $\mathbb{Q}(\sqrt{2})$ parts: per-row TVs, all six pairwise TVs, M_{model} , M_{full} , and NS residuals of the four closed-form witnesses; the saturation check $M_{\text{model}} = 2F^*$ for the enumerated witnesses and the $N = 3$ -NS non-saturation ($M_{\text{model}} = (\sqrt{2} - 1)/2 = \frac{3}{2}F^*$); the converse $F \leq M_{\text{model}} + TV(p_{\text{avg}} \parallel \rho)$ and the fixed-baseline counterexample (all rows $q \neq \rho$ gives $M_{\text{model}} = 0$ with $F > 0$); Hall’s upper bound $S \leq \min\{2 + 3M, 4\}$ re-proved without the no-signalling assumption. **Part G** (high-precision numerics, 60 decimal digits, stated as such): $I(X, Y; \Lambda) = 0.056036$ bits for both $N = 4$ witnesses, 0.054810 for the $N = 3$ NS witness, against the Hall–Branciard floors $I_R \approx 0.046274$ and $I_C \approx 0.080170$ recomputed from their formulas. (The script prints these values to six decimals but asserts them only to 2×10^{-3} ; they were independently recomputed to the printed digits in the review of 2026-09-03.)
- **fringe_pm.py** $\rightarrow$ **fringe_pm.out**, Part PM-7 — Proposition 3 (aggregation robustness): the $N = 4$ witness arithmetic redone in exact $\mathbb{Q}(\sqrt{2})$ — all four per-row TVs equal to $d = (\sqrt{2} - 1)/4$ exactly, $S = 2\sqrt{2}$ exactly, hence the average-TV cost is d ; independent HiGHS LP cross-check minimising the average TV at the known-optimal vertex subject to $S \geq 2\sqrt{2}$ (returns

d to $< 10^{-7}$). The full script contains 88 checks; the remainder certifies the Peres–Mermin results of the companion paper (in preparation).

- **chk2_superdet_garza.py** $\rightarrow$ **out_superdet_garza.txt** — the Garza–Hance figures of Sec. VII D: single-band TV $\sin \theta / (\pi \Delta L)$ (analytic vs numerical integration, $< 1.3 \times 10^{-7}$); the mixture TV $s(\pi - 2s) / (\pi(\pi - s))$ at $\Delta L = 0.77$ (error 1.66×10^{-8} vs numerical integration); self-consistency of their correlator ($\max_{\theta} |E(\theta; \Delta L)| \leq 1$ holds exactly at $\Delta L = \frac{1}{2}$ and fails just below it, 1.0315 at $\Delta L = 0.45$); $|S| \approx 2.82610$ at best fit; the bound $S \leq 2 + 8F$ over full parameter grids (margin $\geq -10^{-9}$). *Flag:* these are floating-point evaluations of the *authors’ published formulas* (the θ values are transcendental, so exact rational/ $\mathbb{Q}(\sqrt{2})$ arithmetic does not apply) — a cross-check of an external model, not a certificate of the theorems of this paper.

3. Independent LP cross-checks (floating point)

- **verify.py** $\rightarrow$ **verify_run.log** (and byte-identical rerun **verify_rerun.log**): 21 checks pass, exit code 0 — the $N = 4$ witness by direct arithmetic; the Bell bound $S_{\max}(F=0) = 2$ for $N = 2, 3, 4$; the curve formula at 12 (N, F) points; F^* bisection for $N = 2, 3, 4$.
- **capregion.out**: the LP reproduces the exact bend values — $S_{\max}(5, \{0.2, 0.25, 0.3\}) = 3.6/3.7/3.8$ ($= \frac{16}{5} + 2F$) to $\leq 1.8 \times 10^{-15}$; $S_{\max}(6, 0.2) = \frac{8}{3} + 4(0.2)$ exactly; the caps $S_{\max}(5, 0.4) = S_{\max}(6, \frac{1}{3}) = 4$ by explicit construction.
- **ns_tight_N{3,4}.out**: direct evaluations $S_{NS}(3, (\sqrt{2} - 1)/3) = S_{NS}(4, (\sqrt{2} - 1)/4) = 2\sqrt{2}$ to 4.4×10^{-16} — the exact Theorem 7 values. Earlier bisection offsets in older logs (≈ 0.13812 , ≈ 0.10358) were resolution artifacts of a 13-iteration bisection on $[0, 0.5]$ (resolution 6.1×10^{-5}), not evidence of an NS gap.
- **ns_witness.out**: the saved NS witnesses re-verified in a fresh process from raw arrays — S to 4.4×10^{-16} , per-row $TV \leq F^* + 10^{-9}$ with at least one row exactly at F^* , all four NS residuals $\leq 5.6 \times 10^{-17}$.

The witness arrays are checked independently of the pipeline that searched for them (recomputation from raw (p, u, v) arrays with plain arithmetic), and the exact certificates are a third route, independent of both.

4. Code availability

The scripts **audit1.py**, **curves_highN.py**, **ns_general.py**, **verify.py**, **prbox_ns.py**, **onesided.py**, **metric_compare.py**, **fringe_pm.py**,

chk2_superdet_garza.py, and the LP pipeline (phase1.py-phase3.py), together with all output logs and witness arrays, the certificate curves_highN.py, and the independent review scripts of 2026-09-03 (indep_check_chsh.py, indep_check_companion.py, cglmp_quantum.py, cglmp_Tprime.py, curves_N9to14.py), are archived in the public repository github.com/sparkdoc/bell-chsh-paper, released with the arXiv posting. Every script runs with python3 from any working directory (NumPy and SciPy are needed only for the floating-point LP cross-checks); each exact certificate exits with code 0 if and only if all of its assertions pass.

Appendix D: Literature sweep

The novelty statements of Sec. VII C are a search result with stated coverage limits, not a claim of

proven novelty. The sweep covered arXiv-indexed literature on measurement dependence and Bell non-locality from 2015 through 2026-09-02 (initial sweep to 2026-08-23; delta check to 2026-09-02 recorded in literature_check2.out, Sec. D): ~ 16 papers were read at full or abstract depth (the rest at catalogue level); pay-walled journal content not mirrored on arXiv, non-arXiv repositories, and books/theses are not covered. The complete record is in literature_check.out, available from the authors on request alongside the other logs: its Section A verifies the anchor works (Hall 2010; Hall & Branciard 2020; Bacciagaluppi, Hermens & Leegwater 2026) and a separate Task-3 section verifies Friedman et al. 2019, all against primary sources; Section B lists the candidate related works found in the sweep with a per-paper overlap verdict (including the four closest: Alai 2026 $\times 2$, Pal et al. 2026, Takakura et al. (arXiv:2208.13634; Quantum 9, 1662 (2025))); Section C states the novelty verdict and these coverage limits; Section D is the search log (queries actually run).

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